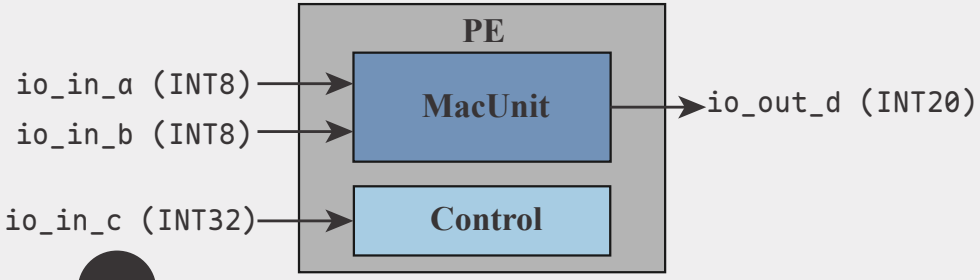
Hardware RTL (SystemVerilog)

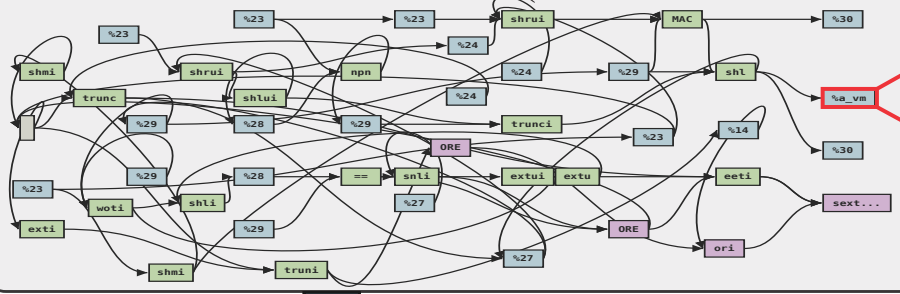



```
// MacUnit.sv - Multiply-accumulate
assign _io_out_d_T = sext(io_in_a) * sext(io_in_b);
assign io_out_d = ... + io_in_c[19:0];

// PE_256.sv - Control flow
always @(posedge clock)
  ... // preload vs MAC result
```

Automated Model Generation

Bit-Level Scalar Model (autoGenILA MLIR)



686 Lines of Low-Level MLIR

```
// The actual multiply
// (buried under 383 ops)
%mul = arith.muli %sext_a, %sext_b : i32
%add = arith.addi %mul, %acc : i32
```

The actual arithmetic (buried)

TensorLift (8-pass semantic lifting)

BitManip → MAC detect → Clamp detect → TAIDL metadata

17 Lines of High-Level MLIR

```
// 8-pass lifting: metadata recovered
%15 = arith.extsi %a : i8 to i32
%16 = arith.muli %15, %14 {tensorlift.mac = 1}
%18 = arith.extsi %17 {tensorlift.clamp}: i16→i20
```

Discovered hardware structure

Recovered saturation behavior